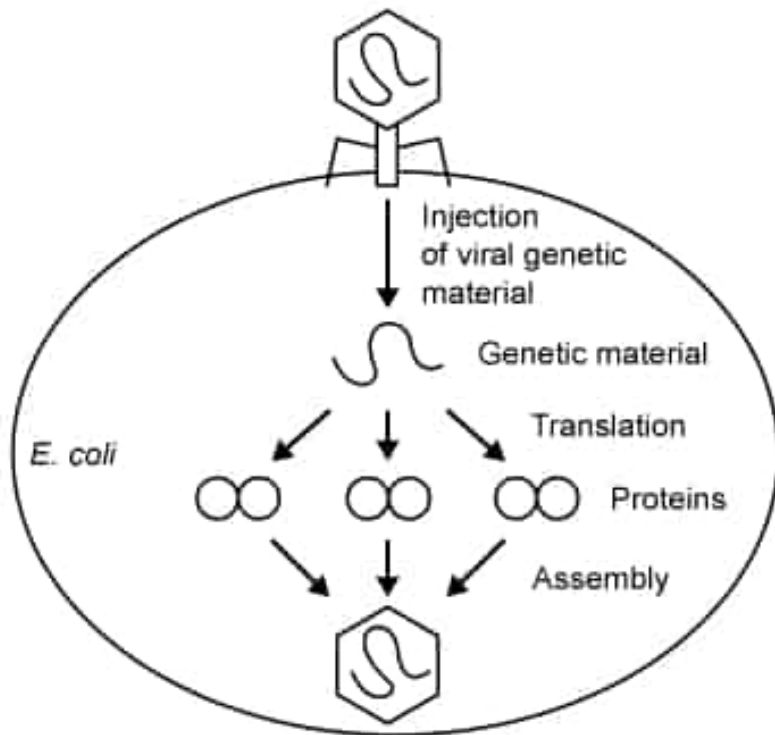


The diagram below shows part of the life cycle of a novel virus that infected an *Escherichia coli* cell.



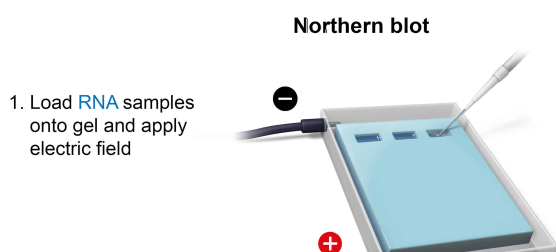
Which of the following experimental techniques can be used to detect the genetic material of this virus in the host cell?

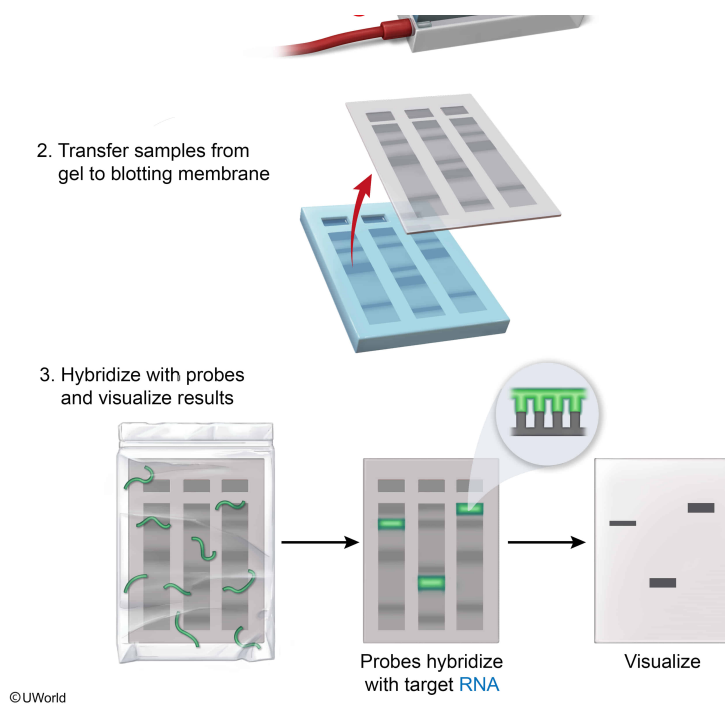
- ☐ A. Southern blot [25%]
- ☐ B. DNA sequencing [13%]
- ✓ ☒ C. Northern blot [57%]
- ☐ D. Gene cloning [3%]

Correct

57% answered correctly

Explanation:





Northern blotting is an assay used to detect and measure the concentration of specific **RNA** molecules within a cell or tissue sample. A typical procedure is as follows:

1. RNA is isolated from the tissue and **denatured**.
2. The RNA is separated by size via **gel electrophoresis**.
3. The RNA samples are transferred from the gel to a blotting membrane and immobilized.
4. The membrane is incubated with a labeled RNA or single-stranded DNA probe complementary to the RNA sequence of interest. If the sequence of interest is present, the probe binds to it (ie, hybridizes) and the label (eg, radioactive atom, fluorescent dye) allows visual detection of the probe.
5. The sample is washed to remove unbound probes.
6. Researchers determine if a visual signal is present. A visual signal indicates that hybridization occurred, meaning the RNA sample contains the RNA sequence of interest. The lack of a visual signal corresponds to an absence of

the target RNA sequence.

Viruses are obligate intracellular parasites with DNA or RNA genomes that can be double-stranded or single-stranded. Specifically, some viruses with positive-sense, single-stranded **RNA genomes** can *directly translate* their genetic material into viral proteins upon infection of a host cell. The given diagram shows that the genome of the novel (ie, newly discovered) virus is *immediately translated* into protein following infection of the *Escherichia coli* cell, with no need for prior **transcription**. Therefore, this virus must have a positive-sense, **single-stranded RNA genome**, which can be detected with **northern blot analysis**.

(Choice A) **Southern blotting** is a technique used to detect and measure the concentration of DNA in a cell. Accordingly, a Southern blot could detect the presence of a viral *DNA* genome but not an RNA genome.

(Choice B) Researchers use DNA sequencing to determine the specific nucleotide order of a DNA molecule. This technique could be used to analyze genome contents, but it is not typically used to *detect the presence* of RNA.

(Choice D) **Gene cloning**, a technique used to generate many copies of a previously identified gene of interest, is not useful for the detection of specific DNA or RNA sequences.

Educational objective:

Northern blotting is a technique used to detect and measure the concentration of a specific RNA sequence in a cell or tissue sample.